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ORIGINAL ARTICLE

A direct polymeric carbon nitride/tungsten oxide Z-scheme heterostructure for efficient photocatalytic hydrogen generation via reforming of plastics into value-added chemicals

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Abstract To promote charge transfer and separation for efficient photocatalysis, a direct Z-scheme heterostructure was constructed by coupling polymeric carbon nitride (PCN) and WO_3. Interestingly, the obtained PCN/WO_3 Z-scheme heterostructure could photocatalytically produce hydrogen and value-added chemicals (e.g., formate and acetate) by reforming of plastic polylactic acid (PLA) in alkaline aqueous condition. By optimizing WO_3 contents and reaction conditions, the obtained PCN/WO_3 heterostructure exhibits much increased photocatalytic activity for PLA photoreforming under visible light, with hydrogen evolution rate reaching 402.90 $\mu mol \cdot g^{-1} \cdot h^{-1}$, which is 3.5 times that of PCN. It is revealed that the Z-scheme charge transfer between PCN and WO_3 mainly contributes to the promoted charge separation and thus the improved photocatalytic activity. Moreover, with h^+ and ·OH experimentally evidenced as the predominant active species, a possible reaction pathway for the reforming of PLA into value-added chemicals (e.g., formate and acetate) over PCN/WO_3 Z-scheme heterostructure is cleared by monitoring the reaction intermediates and radicals. This work paves a carbon neutrality and scalable route toward the synergistical production of hydrogen and value-added chemicals by utilizing and recycling plastic waste.

Keywords Carbon nitride; Z-scheme heterojunction; Plastic waste conversion; Photocatalytic H_2 production; Value-added chemicals

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1 Introduction

To date, more than 8.3 billion metric tons of plastics have been manufactured, and roughly 80% of them have become waste in either landfills or natural environment [1–3]. Inevitably, the ubiquity and non-biodegradability of plastics has resulted in the increased land and ocean pollution, posing a severe threat to our ecosystem [4–6]. Except for landfills and incineration, the current plastic recycling technologies are mainly based on the mechanical processes (e.g., melting and re-extrusion) for recycling-converting from one to another form of plastics, for example, from plastic bottles to fibers. Nevertheless, mechanical recycling could generally produce plastics with poor quality and the times that plastics could be mechanically recycled are also limited [7–9]. Different from mechanical recycling, chemical recycling could depolymerize plastics to produce the monomer raw materials or value-added chemicals. For example, plastic waste could be thermally and thermocatalytically degraded into carbonaceous fuels [10–14]. However, high temperature of up to 300–400 °C and costly metal complex catalysts are always needed to control the product selectivity, which greatly limits their practical applications. Therefore, it is highly desired to develop

novel efficient and sustainable methods to depolymerize and recycle plastics.

As an alternative promising technology, solar-driven photocatalysis has attracted widespread attentions, due to its mild reaction conditions and environmental friendliness in the treatment of polymers (biomacromolecules and plastics) [15–17]. Undoubtedly, to realize efficient photocatalysis, an active, robust and stable semiconducting photocatalyst is of the essence. Till now, various semiconductors have been explored as photocatalysts, including oxides [18–21], sulfides [22–25] and nitrides [26–29]. It is noteworthy that polymeric carbon nitride (PCN) has emerged as a popular photocatalyst in recent years, due to its favorable bandgap (ca. 2.7 eV), low cost, non-toxicity, good stability and thus promising photocatalytic applications [30–34]. However, pristine PCN has suffered from severe charge recombination, resulting in relatively poor photocatalytic performances. To tackle this shortcoming, constructing heterojunctions with suitable band alignment by coupling PCN with a second semiconductor could be effective to improve charge carrier separation and thus enhance photocatalytic performances [35–38]. Typically, some metal oxide semiconductors, e.g., TiO_2 [39] and WO_3 [40], have been explored as the second semiconductor to couple with PCN, with much improved photocatalytic activity achieved over the obtained heterostructures, due to the promoted charge separation.

By deepening the charge transfer processes at the interface of the formed semiconductor/semiconductor heterostructures, the Z-scheme charge transfer process has been believed to favor charge separation and photocatalytic redox reactions [41, 42], under the driving forces of the built-in interfacial electric field, band bending and Coulombic attraction. For example, Zhu et al. successfully developed a PCN/rGO/perylene diimide polymer (PCN/rGO/PDIP) Z-scheme heterostructure for efficient and stable photocatalytic overall water splitting [43]. With a giant internal electric field built to improve the charge carrier separation, the photocatalytic H_2 and O_2 evolution rates could reach as high as 15.80 and 7.80 $\mu mol \cdot h^{-1}$, respectively. Shen et al. reported a two-dimensional PCN-based Z-scheme heterojunction, with band structures of PCN precisely adjusted by the simultaneous introduction of boron dopants and nitrogen defects [44]. Thus, the improved optical absorption ability and the retained high driving forces for water redox reactions greatly benefited photocatalysis for overall water splitting, with solar-to-hydrogen conversion efficiency reaching 1.16% [45].

As inspired by the above demonstration on Z-scheme heterostructures featured with excellent interfacial charge carrier transfer ability, herein, a direct PCN/WO_3 Z-scheme heterostructure was developed for photocatalytic hydrogen production via photoreforming of waste plastics (i.e., polylactic acid, PLA) into value-added chemicals. In comparison with PCN and WO_3, the obtained PCN/WO_3 Z-scheme heterostructure exhibits much improved photocatalytic activity for photoreforming of PLA into value-added chemicals with hydrogen produced simultaneously, under visible light irradiation ($\lambda > 420$ nm). It is well evidenced that the Z-scheme charge transfer process at the interface of PCN and WO_3 could promote charge carrier separation and endow the photoelectrons and photoholes in the PCN/WO_3 heterostructure with high redox driving forces for efficient photocatalytic H_2 production and photoreforming of plastics into value-added chemicals. In addition, with ·OH radicals and holes evidenced as the predominant active species, the reaction pathway for the photoreforming of PLA is reasonably proposed.

2 Experimental

2.1 Materials

All the reagents were of analytical purity and used without further purification. Ammonium tungstate ($(NH_4)_6W_7O_{24}\cdot 6H_2O$, $\geq$ 99.0%), melamine ($C_3H_6N_6$, $\geq$ 99.0%), chloroplatinic acid (H_2PtCl_6, $\geq$ 99.5%), PLA, ethanol (C_2H_6O, $\geq$ 99.5%) were purchased from Sinopharm Chemical Reagent Co., Ltd., China. Ultrahigh purity argon (Ar, $\geq$ 99.999%) was provided by Shaanxi Xinkang Medical Oxygen Co., Ltd. Deionized water, with a resistivity of 18.25 MΩ·cm, was used throughout the experiments.

2.2 Synthesis of PCN

PCN was synthesized according to our previous report [46]. 1 g melamine was placed in crucible with a cover and annealed at 550 °C in air for 4 h with a ramping rate of 5 $°C \cdot min^{-1}$. After naturally cooled down to room temperature, PCN powder in light yellow was obtained.

2.3 Synthesis of WO_3

WO_3 was obtained by annealing $(NH_4)_6W_7O_{24}\cdot 6H_2O$ (1.0 g) in air at 500 °C for 4 h with a ramping rate of 5 $°C \cdot min^{-1}$. After naturally cooled down to room temperature, WO_3 powder was obtained.

2.4 Synthesis of PCN/WO_3-*x* samples

PCN/WO_3-*x* (x = 5, 10, 20) heterostructures were synthesized via annealing the mixture of $(NH_4)_6W_7O_{24}\cdot 6H_2O$ and PCN in air. Typically, different amounts (x mg) of $(NH_4)_6W_7O_{24}\cdot 6H_2O$ (x = 5, 10, 20) were mixed with 200 mg PCN powder, which was then placed in crucible

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with a cover and annealed at 500 °C for 4 h with a ramping rate of 2 °C·min^{-1}. After naturally cooled down to room temperature, the obtained powders were washed with ethanol and deionized water for several times to remove unreacted ammonium ion and dried under vacuum at 80 °C for 10 h. Then, PCN/WO_3-x (x = 5, 10, 20) heterostructures were obtained.

2.5 Material characterization

X-ray diffraction (XRD) patterns were collected on a PANalytical X'pert Pro MPD diffractometer operated at 40 kV and 40 mA using Ni-filtered Cu Kα irradiation (wavelength = 0.15406 nm). Fourier-transform infrared (FTIR) spectra were recorded on a Bruker Vertex 70 FTIR spectrometer using the KBr pellet technique. The amount of the samples used for FTIR was kept at around 1 mg. Scanning electron microscopy (SEM) images were obtained by a JEOL JSM-7800F microscope. Transmission electron microscopy (TEM) images were obtained by a transmission electron microscope (Tecnai G2 F30, FEI) with an accelerating voltage of 300 kV. The elemental compositions and states were recorded by X-ray photoelectron spectroscopy (XPS) with mono aluminum Kα radiation (Axis UltraDLD, Kratos). The charge calibration was done by correcting C 1s line of adventitious carbon setting to 284.8 eV to compensate the charge effect. Ultraviolet–visible–near (UV–Vis) diffuse reflectance spectra (DRS) were recorded on a Cary Series UV–Vis–NIR spectrophotometer (Cary5000, Agilent). Steady-state photoluminescence (PL) emission spectra were carried out at room temperature using a PTI QM-4 fluorescence spectrophotometer. Electron paramagnetic resonance (EPR) experiments were performed on a Bruker EMX X-band spectrometer with frequency = 9.40 GHz at room temperature. For the in situ EPR measurements, 5 mg photocatalyst powders were dispersed in a mixed solution of 5 mL KOH aqueous solution (10 mol·L^{-1}) containing 5 mg PLA and 0.1 mmol 5,5-dimethyl-1-pyrroline N-oxide (DMPO) as the spin-trapping agent, by ultrasonic treatment. Then, the suspension was injected into a glass capillary and then placed in a sealed glass tube under argon (Ar) atmosphere. The sealed glass tube was placed in the microwave cavity of EPR spectrometer and was irradiated with 300 W Xe lamp during EPR measurements at room temperature.

2.6 Hydrolysis treatment of PLA

PLA (25, 50, 125 and 250 mg·mL^{-1}) was hydrolyzed in 10 mol·L^{-1} KOH aqueous solution in a sealed vial for 24 h at 40 °C under stirring at 300 r·min^{-1}. The obtained KOH aqueous solution containing the undissolved pieces of polymer was diluted to 1, 2 and 5 mol·L^{-1} maintaining PLA concentration of 25 mg·mL^{-1} and then used for photoreforming.

2.7 Photocatalytic measurements

The photocatalytic hydrogen evolution reactions were carried out in a hermetic Pyrex reactor. A certain amount of photocatalyst (20 mg) was well dispersed in 30 mL KOH aqueous solution (1, 2, 5, 10 mol·L^{-1}) containing PLA (25 mg·mL^{-1}) as electron donor. 2.0 wt% Pt ($H_2PtCl_6 \cdot 6H_2O$ as the precursor) was loaded on the photocatalyst as a cocatalyst by an in situ photodeposition method. Argon was purged through the reactor for 20 min before reaction to remove the residual air. A 300 W Xe lamp (PLS-SXE300, Beijing PerfectLight) equipped with a 420 nm cutoff filter was used for photocatalytic hydrogen generation. The evolved gases were quantified by a gas chromatography equipped with thermal conductivity detector and 0.5 nm molecular sieve column with argon as carrier gas.

The measurement of apparent quantum yield (AQY) for hydrogen evolution by photoreforming of PLA was measured under the same conditions as for photocatalytic hydrogen evolution with different bandpass filters (λ = 380, 400, 420, 450, 480 and 500 nm) used to provide the monochromatic light. The number of incident photons was determined from a spectroradiometer (AvaSpec-2048-USB2, Netherlands). The AQY value was calculated using Eq. (1) from reported Z-scheme photocatalytic systems [47, 48]:

$$\mathrm{AQY} = \frac{N_{\mathrm{RE}}}{N_{\mathrm{IP}}} \times 100\% = \frac{4 \times N_{\mathrm{E}}}{N_{\mathrm{IP}}} \times 100\% \quad (1)$$

where N_{RE} is the number of reacted electrons, N_{IP} is the number of incident photons, and N_{E} is the number of evolved H_2 molecules. Liquid products were quantified using an high performance liquid chromatography-mass spectrometry (HPLC–MS) (Shimadzu LC–MS 2020) with Wondasil C18 column (5020-39001) and methanol/water = 10:9 (volume: volume) as flow phase.

^{1}H nuclear magnetic resonance (NMR) and ^{13}C NMR spectra were collected on a 400 MHz Bruker spectrometer. All liquid samples were prepared in 2 mol·L^{-1} KOD in D_2O with sample concentrations of 25 mg·mL^{-1}.

2.8 Photoelectrochemical measurements

To prepare working electrode, 1 mg photocatalyst was dispersed in a solution containing H_2O (0.5 mL), EtOH (0.5 mL) and 5 wt% Nafion solution (10 μL), followed by ultrasonication for 30 min. Then, the formed ink (100 μL) was dropped on fluorine-doped tin oxide (FTO) glass

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sheets (glass coated with fluorine-doped tin oxide), which was cleaned with deionized water, acetone and ethanol in sequence. Finally, the working electrode was obtained after naturally dried at room temperature.

Photoelectrochemical characterization was conducted in a standard three-electrode system on an electrochemical workstation (CHI760D), in which the prepared electrode, platinum foil and Ag/AgCl electrode were used as the working, counter and reference electrodes, respectively, with Ar-saturated Na_2SO_4 (0.5 mol·L^{-1}) aqueous solution used as electrolyte. A 500 W Xe lamp (CHF-XM500, Beijing PerfectLight) equipped with an air mass 1.5 global (AM 1.5G) filter was used as light source (100 mW·cm^{-2}). The illumination area of working electrode was fixed at 0.785 cm^2. Electrochemical impedance spectroscopy (EIS) was recorded at an applied potential of 0.8 V vs. Ag/AgCl from 10 kHz to 0.1 Hz. The transient photocurrent response with an interval of 10 s on/off switch was measured at an applied potential of 0.2 V vs. Ag/AgCl. The Mott–Schottky (M-S) plots were measured with 10 mV amplitude at the frequency of 1000 Hz. The measured potentials vs. Ag/AgCl were converted to a reversible hydrogen electrode (RHE) scale according to the Nernst equation [49]:

$$E(\mathrm{RHE}) = E(\mathrm{Ag/AgCl}) + E^0(\mathrm{Ag/AgCl}) + 0.059 \times \mathrm{pH} \quad (2)$$

where E^0(Ag/AgCl) = 0.197 V at 25 °C, E(Ag/AgCl) is the experimentally measured potential against the Ag/AgCl reference electrode, and E(RHE) is the converted potential vs. RHE.

2.9 Density functional theory (DFT) calculations

Spin polarized DFT calculations were conducted using the Vienna ab initio simulation package (VASP) [50] with Perdew–Burke–Ernzerhof (PBE) functional and projector augmented-wave (PAW) method to investigate the work functions of WO_3 (001) and PCN (001). For WO_3, a slab model with five atomic layers and (2 × 1) unit cell (10 W and 30 O atoms) was established. For PCN, a slab model with two atomic layers (24 C and 32 N atoms) was established. To eliminate periodic interference of the unit cell, a vacuum of at least 1.5 nm was added in the Z direction. The convergence tolerances for the electronic structure iteration were set to 1.0 × 10^{-5} eV·atom^{-1} for energy and 0.1 eV·nm^{-1} for maximum force. The cutoff energy was set as 500 eV after testing. The Monkhorst–Pack grid of k-point was set as 3 × 3 × 1 and 4 × 2 × 1 for WO_3 and PCN models. The DFT-D3 method of Grimme et al. [51] was used for the vdW-dispersion energy correction.

3 Results and discussion

3.1 Chemical and electronic structures

Herein, PCN/WO_3 heterostructures with different WO_3 contents were prepared by a simple two-step solid-state annealing method (Fig. 1a, as seen in experimental section). Typically, PCN with lamellar structure (Fig. 1b), obtained by annealing melamine at 823 K for 4 h, was mixed with different amounts (x mg, x = 5, 10, 20) of $(NH_4)_6W_7O_{24}\cdot 6H_2O$ and then annealed at 773 K for 4 h. In the obtained PCN/WO_3 heterostructures (PCN/WO_3-x, x = 5, 10, 20), WO_3 nanoparticles could be observed at the surface of PCN (Fig. 1c, taking PCN/WO_3-10 as an example), as identified by the (020) plane of monoclinic WO_3 with lattice spacing of 0.376 nm (Fig. 1d), matching well with the lattice structure of WO_3 nanoparticles obtained by thermal annealing (Fig. S1, as seen in experimental section). As observed in energy-dispersive X-ray spectroscopy (EDS) elemental mappings (Fig. S2), the homogeneous distribution of C and N indicates the presence of PCN, while the distribution of O and W spatially different from those of C and N illustrates the successful decoration of WO_3 nanoparticles in the formed PCN/WO_3 heterostructures.

The crystal structures of the obtained PCN/WO_3-x heterostructures were investigated by XRD. Depending on the increasing WO_3 contents, XRD peaks (Fig. 1e) assigned to hexagonal WO_3 observed at 2θ = 23.1°, 23.7°, 24.4°, 28.1° and 36.8° become more intensive, with the characteristic XRD peaks at 2θ = 13.1° and 27.4° well maintained for PCN, which indicates the formation of PCN/WO_3-x heterostructures with the crystal structure of PCN unchanged. The chemical structure evolution depending on the increasing WO_3 contents was then monitored by FTIR spectra. It is clear that all the PCN/WO_3-x heterostructures present FTIR spectra similar to the pristine PCN (Fig. 1f), with a broad absorption band observed in the range of 3000–3400 cm^{-1}, belonging to uncondensed terminal amino groups (-NH_2 or -NH groups) [52]. The multiple bands in the range of 1250–1700 cm^{-1} could be assigned to the stretching vibrations of aromatic C-N heterocycles containing either trigonal N(-C)$_3$ or bridging C-NH-C units, while the sharp peak at ca. 815 cm^{-1} could be attributed to the out-of-plane bending mode of heptazine rings. It is also observable that depending on the increased WO_3 contents, an additional peak gradually emerges in the range of 500–1000 cm^{-1}, related to the O–W–O stretching absorption [53], evidencing the formation of PCN/WO_3 heterostructures. However, with the increasing WO_3 contents, FTIR bands assigned to the PCN heterocycles and the tri-s-triazine ring structure get gradually weakened. This

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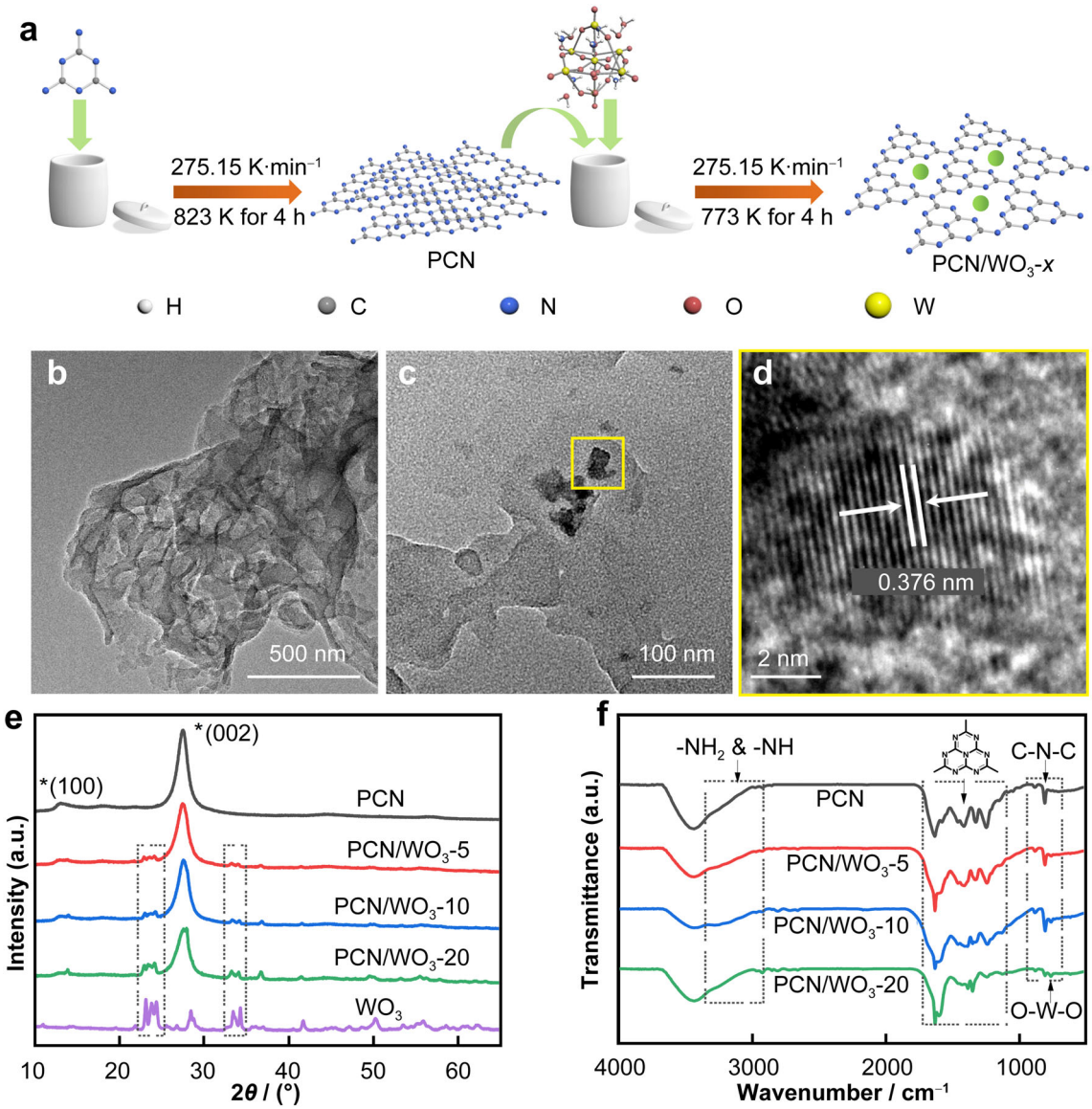

Fig. 1 **a** Schemed synthesis process of PCN/WO_3-*x* heterostructures; TEM images of **b** PCN and **c, d** PCN/WO_3-10; **e** XRD patterns of WO_3, PCN and PCN/WO_3-*x*; and **f** FTIR spectra of PCN and PCN/WO_3-*x*

molecular change could be explained by the partial destruction of PCN molecular structure, induced by the chemical interaction between PCN and WO_3 during thermal annealing [54].

XPS measurements were conducted to investigate the chemical compositions and elemental coordination of PCN/WO_3-*x* (taking PCN/WO_3-10 as the example). Pristine PCN and PCN/WO_3-10 exhibit similar C 1s XPS spectra (Fig. 2a), which could be deconvoluted into three peaks located at ca. 288.5, 286.2 and 284.8 eV, respectively, attributed to the sp^2-bonded carbon (N–C=N) in the framework of PCN, the sp^3-coordinated carbon (C−NH_2) on the edges of heptazine units and the sp^2 C−C/C=C bonds of graphitic carbon. Note the positive shift of ca. 0.3 eV occurring in the binding energy of C 1s peak assigned to the N–C=N for PCN/WO_3-10 as compared to PCN, the electron transfer from PCN to WO_3 happens in the obtained PCN/WO_3-10 heterostructure. For N 1s XPS peaks located at ca. 401.1, 399.5 and 398.5 eV (Fig. 2b), corresponding to the amino groups (C−NH_2), the tertiary nitrogen groups (N−$(C)_3$) and the sp^2-bonded N (C−N=C), respectively, a positive shift of ca. 0.2 eV in binding energy could be observed for N−$(C)_3$ and C−N=C of PCN/WO_3-10 as compared to PCN, as a result of the electron transfer from PCN to WO_3. W 4f levels for both WO_3 and PCN/WO_3-10 (Fig. 2c) are split into two peaks of W $4f_{7/2}$ and W $4f_{5/2}$ with binding energy of ca. 35.5 and 37.5 eV, respectively, which corresponds to the W^{6+} state.

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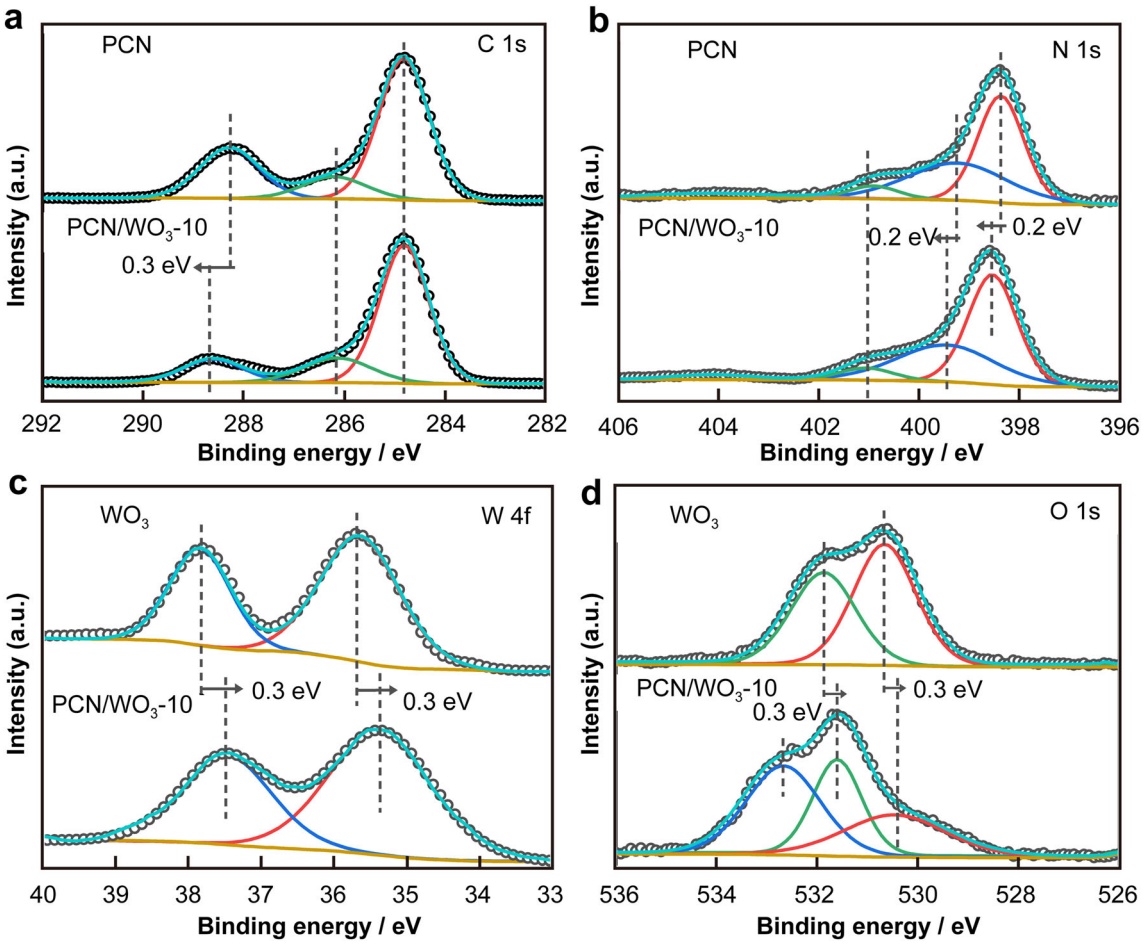

Fig. 2 **a** C 1s, **b** N 1s, **c** W 4f and **d** O 1s XPS spectra of PCN/WO_3-10 and PCN or WO_3 as reference

One would note that these two peaks are negatively shifted by ca. 0.3 eV for PCN/WO_3-10, as compared to WO_3, again implying the electron transfer behavior in the formed heterostructure [42]. The above XPS analysis evidences the formation of PCN/WO_3 heterostructures by two-step solid-state annealing method, with electronic interaction happening between PCN and WO_3, which should be responsible for the realization of interfacial Z-scheme transfer charge process. For the O 1s XPS spectra (Fig. 2d), two XPS peaks could be observed at ca. 530.7 and 531.9 eV, respectively, which could be assigned to the lattice oxygen of WO_3 and hydroxyl oxygen. In comparison with WO_3, the 0.3 eV negative shift at XPS peaks of lattice oxygen happening for PCN/WO_3-10 indicates the electron transfer from PCN to WO_3. The much increased intensities of XPS peaks assigned to hydroxyl oxygen at 531.6 eV for PCN/WO_3-10 relative to WO_3 suggest that PCN is beneficial for the surface adsorption of OH groups, which would provide highly accessible active sites for PLA and accelerating photogenerated electron–hole pair separation and charge transfer [55–57]. An additional peak is observed for PCN/WO_3-10 at 532.6 eV, which should correspond to the oxygen atoms in chemically adsorbed H_2O, attributed to the high content of PCN in the heterostructure as evidenced by XRD results. Such interaction between water molecules and photocatalysts at surface would further benefit photogenerated charge transfer and surface water redox reactions [58].

3.2 Photocatalytic performance

The photocatalytic performances for hydrogen evolution over PCN/WO_3-x heterostructures were evaluated in KOH aqueous solution containing hydrolysis-treated plastic PLA under visible light irradiation ($\lambda > 420$ nm) (as seen in experimental section). With the conduction band potential of WO_3 more positive than the water reduction potential [59], WO_3 could hardly photocatalytically produce hydrogen from water. With WO_3 introduced into PCN, all the PCN/WO_3-x heterostructures exhibit much enhanced photocatalytic hydrogen evolution performances (Fig. 3a), with the highest hydrogen evolution rate reaching 284.49 $\mu mol \cdot g^{-1} \cdot h^{-1}$ for PCN/WO_3-10, which is about 2.5 times that of pure PCN. As well documented that the

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alkaline concentrations of photocatalytic reaction solution would be greatly determinative to the photocatalytic activity for high evolution via photoreforming of plastics [60]. Herein, the concentrations of KOH solutions were further optimized to improve the photocatalytic activity of PCN/WO_3-10 for hydrogen evolution. As shown in Fig. 3b, the photocatalytic hydrogen activity could be significantly with KOH concentrations increasing from 1 to 2 mol·L^{-1}, maybe due to promoted dissolution of plastic molecules producing abundant active substances. However, a further increase in KOH concentration may destroy the molecular and crystal structure of both components in PCN/WO_3-10, giving rise to decreased photocatalytic activity for hydrogen evolution. With WO_3 contents and KOH concentrations optimized, encouragingly, PCN/WO_3-10 exhibits excellent photocatalytic activity for hydrogen generation via photoreforming of PLA, with hydrogen generation rate reaching 402.90 μmol·g^{-1}·h^{-1}, which surpasses most of the reported photocatalysts (Table S1). The close dependence of AQY values on the optical absorption profile indicates that the hydrogen production is indeed a photocatalysis process, with AQY calculated to be as high as 1.36% at 420 nm (Fig. S3). One would also note that the obtained PCN/WO_3-10 heterostructure reveals good photocatalytic stability, with hydrogen evolution rates well maintained during a 24-h reaction (Fig. 3c). The almost unchanged XRD patterns before and after photocatalytic reaction (Fig. S4) also evidence the good stability of PCN/WO_3-10 for photocatalytic hydrogen generation via reforming of plastic PLA in alkaline solution. As expected, along with hydrogen production, the photoreforming of PLA would produce value-added chemicals, with formate and acetate detected as the value-added products in the reaction solution (Fig. 3d).

It is noted that in comparison with the pristine PCN, PCN/WO_3-10 exhibits much increased photocatalytic activity for hydrogen evolution and photoreforming of PLA with value-added chemicals produced (Fig. 3). To reveal the determinants of the improved photocatalytic activity achieved over the PCN/WO_3-x heterostructures, the optical

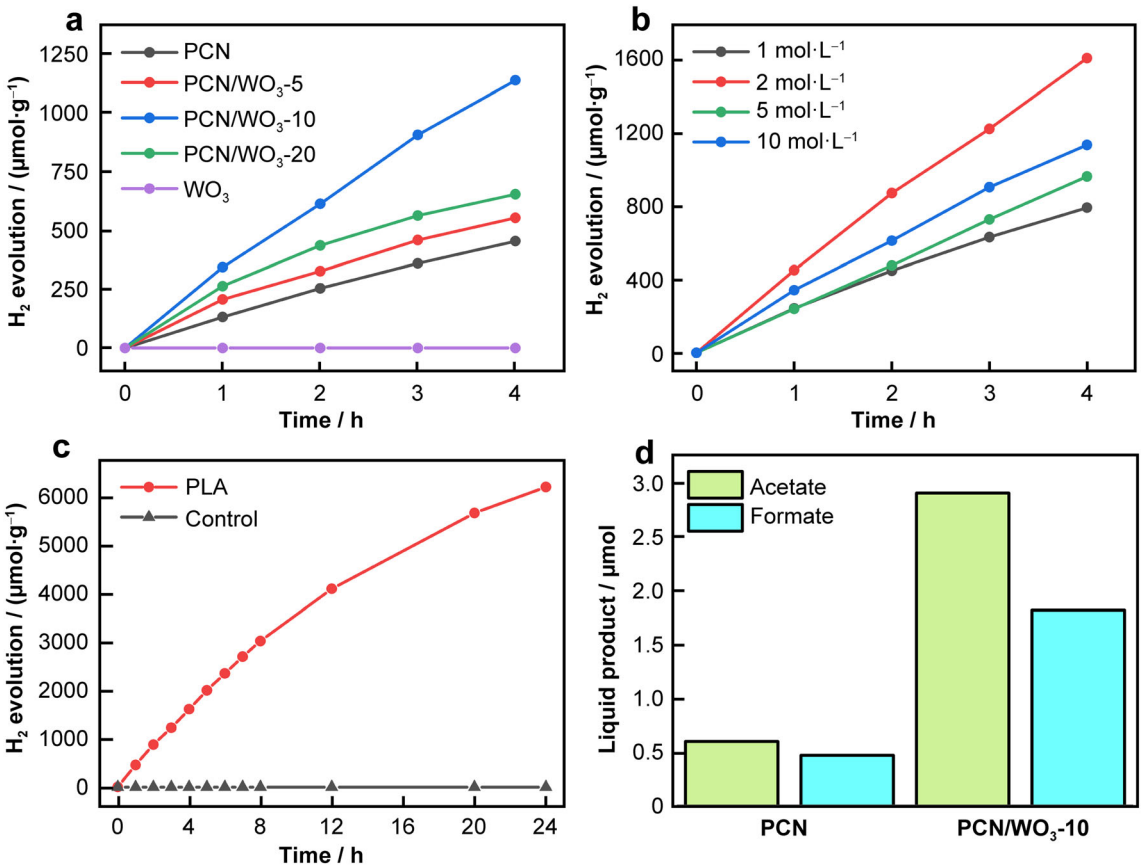

Fig. 3 Visible light ($\lambda > 420$ nm)-driven photocatalytic hydrogen production by photoreforming of PLA over **a** PCN/WO_3-x in KOH solution (10 mol·L^{-1}) and **b** PCN/WO_3-10 in KOH solutions with different concentrations (1, 2, 5, 10 mol·L^{-1}); **c** long-term photocatalytic hydrogen production over PCN/WO_3-10 by photoreforming of PLA in KOH solution (2 mol·L^{-1}) under visible light irradiation ($\lambda > 420$ nm) with controlled experimental condition: in dark conditions with photocatalyst or under irradiation without photocatalyst. **d** Amounts of value-added chemicals produced during 24 h photoreforming of PLA in 2 mol·L^{-1} KOH aqueous solution

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and electrochemical properties were systematically investigated. Note the very similar optical absorption profiles for WO_3, PCN and PCN/WO_3-x as revealed by UV–Vis–NIR DRS (Fig. S5), the optical absorption property should not be determinative to the much increased photocatalytic activity. PL spectra, originating from the radiative recombination of electrons and holes [61], were then collected to evaluate charge carrier transfer ability in the obtained PCN/WO_3-x heterostructures. Both pristine PCN and PCN/WO_3-x heterostructures show significant PL emission peaks at 445 nm (Fig. 4a), close to the absorption edge of pristine PCN. It is then observed that all the PCN/WO_3-x heterostructures exhibit quenched PL intensity as compared to PCN, and especially PCN/WO_3-10 shows the lowest PL intensity, which indicates that appropriate introduction of WO_3 into PCN would effectively inhibit charge carrier recombination. Thus, the effective separation of photoexcited charge carriers should contribute to the excellent photocatalytic activity of PCN/WO_3-10, which could be again corroborated by EPR spectra (Fig. 4b). With EPR signal at g factor of ca. 2.003 assigned to the unpaired electrons of oxygen defects in the WO_3 structure hardly detected for WO_3 [62], the single Lorentzian line centered at g = 2.0034 observed for both PCN and PCN/WO_3-10 should be then attributed to the unpaired electrons in the aromatic rings of carbon atoms in PCN. EPR intensity of PCN/WO_3-10 is greatly enhanced relative to that of PCN, which suggests that the introduction of WO_3 could enhance the π system delocalization and improve the electron mobility [63]. Additional evidence supporting the promoted charge carrier transfer process in the PCN/WO_3 heterostructures could be provided by EIS measurements (Fig. 4c). As informed from the Nyquist plots fitted by a resistor–capacitance (RC) equivalent circuit (Table S2, inset in Fig. 4c), the bulk charge transport resistance (R_{bulk}) observed for PCN/WO_3-10 (523.5 Ω) is lower than those for PCN (788.5 Ω) and WO_3 (688.5 Ω), which indicates that introduction of WO_3 into PCN could accelerate the charge transport process in the formed PCN/WO_3-10 heterostructure. Furthermore, the interfacial charge transport resistance (R_{ct}) is much decreased for PCN/WO_3-10 (831 Ω) as compared to PCN (2871 Ω) and WO_3 (1428 Ω), indicating the accelerated charge transfer process at the interface of PCN/WO_3-10 and electrolyte.

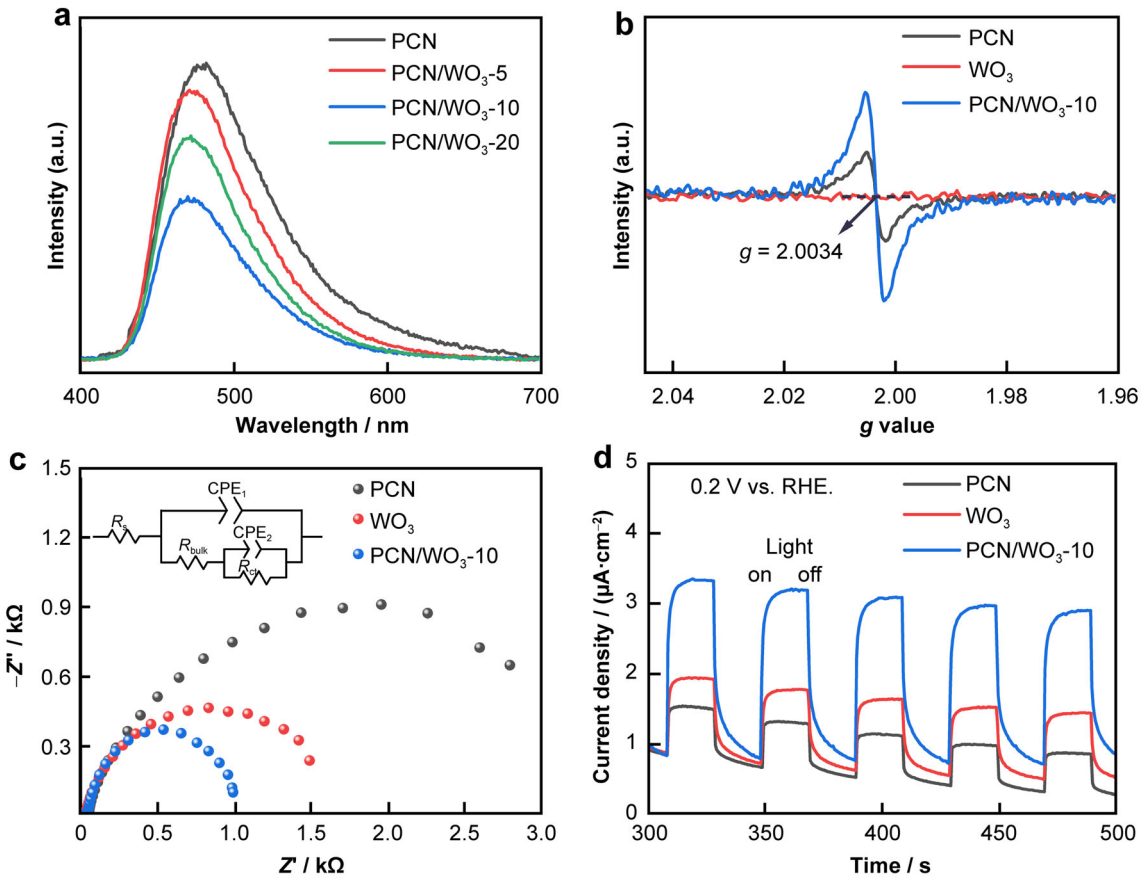

Fig. 4 **a** Steady-state PL spectra of PCN/WO_3-x; **b** EPR spectra, **c** EIS Nyquist plots and **d** transient photocurrent responses of PCN, WO_3 and PCN/WO_3-10 and (inset) equivalent circuit

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This significant improvement in charge transfer ability in the bulk of PCN/WO_3-10 and at the photocatalyst/electrolyte interface could be further supported by the transient photocurrent response properties (Fig. 4d). In comparison with PCN and WO_3, PCN/WO_3-10 exhibits much increased photocurrent density, again proving the significantly promoted charge carrier separation in the PCN/WO_3-10 heterostructures, which should contribute to the remarkably improved photocatalytic performances.

3.3 Photocatalytic mechanism

To deepen the understanding of charge transfer processes contributing to the improved photocatalytic activity of the PCN/WO_3-*x* heterostructures, the band structures were optically and electrochemically determined, with reaction pathways proposed for the photoreforming of PLA. As calculated by the Kubelka–Munk method (Fig. S6), the band gap (E_g) values are determined to be 2.74 and 2.48 eV for PCN and WO_3, respectively. Mott–Schottky (M–S) plots (Fig. S7) exhibit positive slopes for both PCN and WO_3, indicating their n-type semiconductor characteristics [64, 65], with conduction band (CB) positions determined to be located at −1.38 and −0.29 V vs. Ag/AgCl, i.e., −0.77 and + 0.32 V vs. RHE, respectively. Then, the valence band (VB) positions of PCN and WO_3 are calculated to be 1.97 and 2.80 V vs. RHE, respectively. Given the well-determined band structures of PCN and WO_3, the Z-scheme charge transfer process in the formed PCN/WO_3-*x* could be reasonably proposed (Fig. 5a). With work functions theoretically calculated to be 6.44 and 4.74 eV for PCN and WO_3 (Figs. S8 and S9a), respectively, it could be expected that electrons will spontaneously flow from PCN to WO_3 to rebalance the Fermi levels, when PCN and WO_3 are in close contact. Consequently, the positively charged PCN and the negatively charged WO_3 would create an internal electric field directed from PCN to WO_3 at the interface (Fig. S9b) [37]. As electrons in PCN experience repulsion from the internal electric field, the potential energy of electrons increases and the bands bend upwards, and conversely, the potential energy of electrons in WO_3 decreases and the bands bend downwards (Fig. S9c) [66]. As driven by the internal electric field and the interfacial band bending, the photogenerated electrons at the CB of WO_3 are likely to transfer through the interface to recombine with the holes at the VB of PCN, which suggests a Z-scheme charge transfer process in the PCN/WO_3 heterostructure (Fig. S9d).

EPR spin-trapping measurements were then conducted to further verify the Z-scheme charge transfer process in PCN/WO_3, with DMPO deployed to trap radicals (e.g., ·OH and $\cdot O_2^-$) in methanol and aqueous solution. Under illumination, the DMPO-·OH signals could be observed for WO_3 and PCN/WO_3-10 in aqueous solution, while no DMPO-·OH signal is observed for PCN (Fig. 5b). This phenomenon is explainable, as the holes in the VB (1.97 V vs. RHE) of PCN could hardly oxidize H_2O to ·OH (2.40 V vs. RHE), while the potential of the holes in the VB of WO_3 (2.80 vs. RHE) is positive enough to produce ·OH. Moreover, strong DMPO-$\cdot O_2^-$ signals could be observed for PCN and PCN/WO_3-10 samples in methanol solution (Fig. 5c), implying that the photogenerated electrons in PCN and PCN/WO_3-10 have enough reduction ability to reduce O_2 to $\cdot O_2^-$. However, no obvious DMPO-$\cdot O_2^-$ signal is observed for WO_3, indicating that the electrons in the CB of WO_3 could hardly reduce O_2 to $\cdot O_2^-$, as the CB potential (0.32 V) is more positive than the redox potential of O_2/$\cdot O_2^-$ (− 0.33 V vs. RHE). Interestingly, PCN/WO_3-10 exhibits much stronger signals than PCN or WO_3 in both aqueous and methanol solutions. These EPR results evidence that the photoexcited electrons and holes with high redox driving forces are well maintained in the CB of PCN and the VB of WO_3, respectively, in the formed PCN/WO_3 heterostructures. Therefore, rather than the conventional Type II heterostructure with photoexcited electrons and holes accumulated at the CB of WO_3 and the VB of PCN (Fig. S9c), respectively, the Z-scheme charge transfer process (Figs. 5a and S9d) is realized in the formed PCN/WO_3 heterostructure, which could well explain the enhanced photocatalytic activity for hydrogen evolution and photoreforming of PLA.

For the photoreforming of PLA, it should be mentioned that KOH pretreatment (as seen in experimental section) could hydrolyze PLA into lactic acid (LA) (Fig. S10a, b). After 120 h photoreforming of hydrolysis-pretreated PLA solution, value-added chemicals, such as formate and acetate, are produced (Fig. S10c, d), as verified by authentic molecules (Fig. S11). With DMPO added into the hydrolysis-treated PLA solution, one could observe six weak EPR peaks with intensity ratio of 1:1:1:1:1:1 for PCN under visible light irradiation (Fig. 5d), which should be assigned to the carbon-centered radicals. This EPR signal is much enhanced for PCN/WO_3-10 as compared with PCN, indicating that more carbon-centered radicals are generated via photoreforming of PLA over PCN/WO_3-10. Except for the EPR signals related to carbon-centered radicals, the four-peak EPR signal assigned to ·OH radicals, with intensity ratio of 1:2:2:1, could be detected for PCN/WO_3-10. It should be also noted that the EPR signal of ·OH radicals is stronger than that of carbon-center radicals. These observations evidence that PCN/WO_3-10 is more likely to generate highly reactive ·OH radicals, which would promote the cleavage of the α-H bonds in LA and then accelerate the conversion of LA to acetate and formate [67, 68]. Given the detected EPR signals of ·OH radicals after short time irradiation (Fig. S12) and

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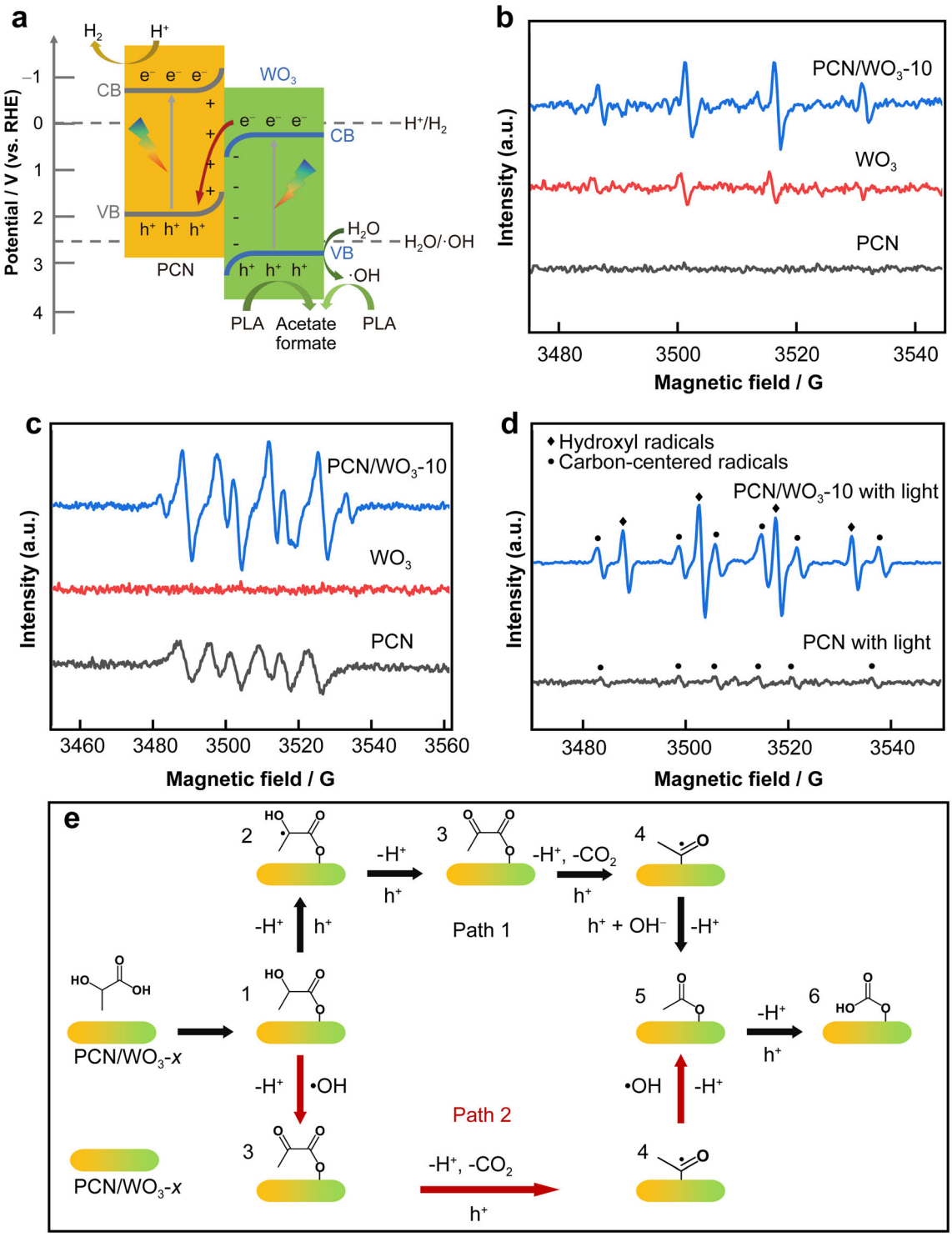

Fig. 5 **a** Diagram of Z-scheme charge transfer processes in the PCN/WO_3-*x* heterostructure; EPR spectra of **b** DMPO-·OH in aqueous solution and **c** DMPO-·O_2^- in methanol solution in presence of PCN, WO_3 and PCN/WO_3-10 under visible light irradiation; **d** in situ EPR spectra of hydrolysis-treated PLA solution with DMPO added as ·OH scavenger in presence of PCN and PCN/WO_3-10 under visible light irradiation; and **e** proposed reaction pathways for photoreforming of PLA

the greatly inhibited hydrogen production with DMPO added (Fig. S13) over PCN/WO_3-10, it could be further confirmed that ·OH should contribute to the production of value-added chemicals via LA oxidation. Given the well-detected photoreforming products (Fig. S10c, d) and reaction radicals (Fig. 5d), the reaction pathways for the

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photoreforming of PLA could be reasonably proposed by following two paths (Fig. 5e), after PLA is hydrolyzed to LA (1) (Fig. S10a, b). For Path 1, LA (1) is oxidized by holes retained in PCN/WO_3, which is then deprotonated to produce carbon-centered lactate radicals (2) and further pyruvate (3) [67]. In the following, decarboxylation and deprotonation happen to pyruvate (3), producing acetaldehyde intermediates (4), which is subsequently oxidized by holes in alkaline condition to obtain acetate (5) (Fig. S10c, d). Further α-H oxidation of acetate (5) would produce formate (6) (Fig. S10c, d). For Path 2, one should also note that ·OH radicals could trigger dehydration of LA (1) to directly produce pyruvate (3), which is then easily oxidatively decarboxylated, due to the electron withdrawing effect of the α-carbonyl group. The decarboxylated acetaldehyde intermediates (4) are further oxidized by ·OH radicals in alkaline condition to obtain acetate (5). Via α-H oxidation, acetate is then converted to formate (6).

4 Conclusion

In summary, a PCN/WO_3 heterostructure featured with the Z-scheme charge transfer process was synthesized by a two-step solid-state annealing approach. With formate and acetate detected as value-added chemicals produced by photoreforming of PLA, the obtained PCN/WO_3 heterostructure exhibits a significant improvement in photocatalytic performance under visible light ($\lambda > 420$ nm), with the highest hydrogen evolution rate reaching 402.90 $\mu mol \cdot g^{-1} \cdot h^{-1}$, about 3.5 times than that of PCN. It is revealed that the improved photocatalytic activity should be ascribed to the formation of direct Z-scheme heterojunction, with high redox driving forces well retained for the photogenerated electrons and holes to get involved in hydrogen evolution reaction and photoreforming of PLA, respectively. By monitoring the reaction intermediates and radicals, the reaction pathway for the photoreforming of PLA is reasonably proposed, with ·OH radicals and holes acting as the predominant active species. This study demonstrates a facile strategy to delicately design heterostructured photocatalysts for efficient solar hydrogen conversion, with value-added chemicals simultaneously produced by recycling of plastic waste.

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Declarations

Conflict of interests The authors declare that they have no conflict of interest.

References

[1] Gibb BC. Plastics are forever. Nat Chem. 2019;11(5):394. https://doi.org/10.1038/s41557-019-0260-7.
[2] Geyer R, Jambeck JR, Law KL. Production, use, and fate of all plastics ever made. Sci Adv. 2017;3(7):e1700782. https://doi.org/10.1126/sciadv.1700782.
[3] Garcia JM, Robertson ML. The future of plastics recycling. Science. 2017;358(6365):870. https://doi.org/10.1126/science.aaq0324.
[4] Jambeck JR, Geyer R, Wilcox C, Siegler TR, Perryman M, Andrady A, Narayan R, Law KL. Plastic waste inputs from land into the ocean. Science. 2015;347(6223):768. https://doi.org/10.1126/science.1260352.
[5] Sardon H, Dove AP. Plastics recycling with a difference. Science. 2018;360(6387):380. https://doi.org/10.1126/science.aat4997.
[6] Tournier V, Topham CM, Gilles A, David B, Folgoas C, Moya-Leclair E, Kamionka E, Desrousseaux ML, Texier H, Gavalda S, Cot M, Guémard E, Dalibey M, Nomme J, Cioci G, Barbe S, Chateau M, André I, Duquesne S, Marty A. An engineered PET depolymerase to break down and recycle plastic bottles. Nature. 2020;580(7802):216. https://doi.org/10.1038/s41586-020-2149-4.
[7] Ragaert K, Delva L, Van Geem K. Mechanical and chemical recycling of solid plastic waste. Waste Manage. 2017;69:24. https://doi.org/10.1016/j.wasman.2017.07.044.
[8] Jehanno C, Alty JW, Roosen M, De Meester S, Dove AP, Chen EYX, Leibfarth FA, Sardon H. Critical advances and future opportunities in upcycling commodity polymers. Nature. 2022;603(7903):803. https://doi.org/10.1038/s41586-021-04350-0.
[9] Suzuki G, Uchida N, Tuyen LH, Tanaka K, Matsukami H, Kunisue T, Takahashi S, Viet PH, Kuramochi H, Osako M. Mechanical recycling of plastic waste as a point source of microplastic pollution. Environ Pollut. 2022;303:119114. https://doi.org/10.1016/j.envpol.2022.119114.
[10] Wong S, Ngadi N, Tuan Abdullah TA, Inuwa IM. Catalytic cracking of LDPE dissolved in benzene using nickel-impregnated zeolites. Ind Eng Chem Res. 2016;55(9):2543. https://doi.org/10.1021/acs.iecr.5b04518.
[11] Burange AS, Gawande MB, Lam FLY, Jayaram RV, Luque R. Heterogeneously catalyzed strategies for the deconstruction of high density polyethylene: plastic waste valorisation to fuels. Green Chem. 2015;17(1):146. https://doi.org/10.1039/C4GC01760A.
[12] Miandad R, Barakat MA, Aburiazaiza AS, Rehan M, Nizami AS. Catalytic pyrolysis of plastic waste: a review. Process Saf Environ Prot. 2016;102:822. https://doi.org/10.1016/j.psep.2016.06.022.
[13] Vance BC, Kots PA, Wang C, Hinton ZR, Quinn CM, Epps TH, Korley LTJ, Vlachos DG. Single pot catalyst strategy to branched products via adhesive isomerization and hydrocracking of polyethylene over platinum tungstated zirconia. Appl Catal B. 2021;299:120483. https://doi.org/10.1016/j.apcatb.2021.120483.
[14] Vogt BD, Stokes KK, Kumar SK. Why is recycling of post-consumer plastics so challenging? ACS Appl Polym Mater. 2021;3(9):4325. https://doi.org/10.1021/acsapm.1c00648.
[15] Wu X, Xie S, Zhang H, Zhang Q, Sels BF, Wang Y. Metal sulfide photocatalysts for lignocellulose valorization. Adv Mater. 2021;33(50):2007129. https://doi.org/10.1002/adma.202007129.
[16] Pichler CM, Bhattacharjee S, Rahaman M, Uekert T, Reisner E. Conversion of polyethylene waste into gaseous hydrocarbons via integrated tandem chemical–photo/electrocatalytic processes.

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ACS Catal. 2021;11(15):9159. https://doi.org/10.1021/acscatal.1c02133.

[17] Zhou C, Wu J, Li Y, Cao H. Highly efficient UV-visible-infrared light-driven photothermocatalytic steam biomass reforming to H_2 on Ni nanoparticles loaded on mesoporous silica. Energy Environ Sci. 2022;15(7):3041. https://doi.org/10.1039/D2EE00816E.

[18] Costa MB, de Araújo MA, Tinoco MVDL, Brito JFD, Mascaro LH. Current trending and beyond for solar-driven water splitting reaction on WO_3 photoanodes. J Energy Chem. 2022;73:88. https://doi.org/10.1016/j.jechem.2022.06.003.

[19] Wang Q, Cao XX, Liu T, Wu KJ, Deng J, Chen JS, Cai YJ, Shen MQ, Yu C, Wang WK. Microfluidic assembly of WO_3/MoS_2 Z-scheme heterojunction as tandem photocatalyst for nitrobenzene hydrogenation. Rare Met. 2023;42(2):484. https://doi.org/10.1007/s12598-022-02169-w.

[20] Cai L, Du Y, Guan X, Shen S. CdS nanocrystallites sensitized ZnO nanorods with plasmon enhanced photoelectrochemical performance. Chin Chem Lett. 2019;30(12):2363. https://doi.org/10.1016/j.cclet.2019.07.020.

[21] Wei D, Tan Y, Wang Y, Kong T, Shen S, Mao SS. Function-switchable metal/semiconductor junction enables efficient photocatalytic overall water splitting with selective water oxidation products. Sci Bull. 2020;65(16):1389. https://doi.org/10.1016/j.scib.2020.04.042.

[22] Wang L, Song XY, Ong HJ, Song YN. Preparation of flower--like Bi/CuS photocatalyst by photo-reduction method. Chin J Rare Met. 2023;47(1):177. https://doi.org/10.13373/j.cnki.cjrm.XY22060030.

[23] Qin Z, Guan X, Guo X, Guo P, Wang M, Huang Z, Chen Y. Integrated Z-scheme nanosystem based on metal sulfide nanorods for efficient photocatalytic pure water splitting. Chemsuschem. 2020;13(24):6528. https://doi.org/10.1002/cssc.202002171.

[24] Liu F, Xue F, Zhang C, Lu K, Chen G, Ye X, Chen W, Guo L, Liu M. Noble-metal-free chalcogenide nanotwins for efficient and stable photocatalytic pure water splitting by surface phosphorization and cocatalyst modification. Mater Today Energy. 2022;30:101180. https://doi.org/10.1016/j.mtener.2022.101180.

[25] Liu F, Xue F, Si Y, Chen G, Guan X, Lu K, Liu M. Functionalized $Cd_{0.5}Zn_{0.5}S$ chalcogenide nanotwins enabling Z--scheme photocatalytic water splitting. ACS Appl Nano Mater. 2021;4(1):759. https://doi.org/10.1021/acsanm.0c03054.

[26] Chen S, Qi Y, Hisatomi T, Ding Q, Asai T, Li Z, Ma SSK, Zhang F, Domen K, Li C. Efficient visible-light-driven Z--scheme overall water splitting using a $MgTa_2O_{6-x}N_y$/TaON heterostructure photocatalyst for H_2 evolution. Angew Chem Int Ed. 2015;54(29):8498. https://doi.org/10.1002/anie.201502686.

[27] Cui J, Liu T, Dong B, Qi Y, Yuan H, Gao J, Yang D, Zhang F. Flux-assisted synthesis of prism-like octahedral Ta_3N_5 single-crystals with controllable facets for promoted photocatalytic H_2 evolution. Solar RRL. 2021;5(6):2000574. https://doi.org/10.1002/solr.202000574.

[28] Nurlaela E, Ould-Chikh S, Llorens I, Hazemann JL, Takanabe K. Establishing efficient cobalt-based catalytic sites for oxygen evolution on a Ta_3N_5 photocatalyst. Chem Mater. 2015;27(16):5685. https://doi.org/10.1021/acs.chemmater.5b02139.

[29] Li Y, Wang Y, Dong CL, Huang YC, Chen J, Zhang Z, Meng F, Zhang Q, Huangfu Y, Zhao D, Gu L, Shen S. Single-atom nickel terminating sp^2 and sp^3 nitride in polymeric carbon nitride for visible-light photocatalytic overall water splitting. Chem Sci. 2021;12(10):3633. https://doi.org/10.1039/D0SC07093A.

[30] Zhang Y, Huang Z, Dong CL, Shi J, Cheng C, Guan X, Zong S, Luo B, Cheng Z, Wei D, Huang YC, Shen S, Guo L. Synergistic effect of nitrogen vacancy on ultrathin graphitic carbon nitride porous nanosheets for highly efficient photocatalytic H_2 evolution. Chem Eng J. 2022;431:134101. https://doi.org/10.1016/j.cej.2021.134101.

[31] Lin Z, Zhang Z, Wang Y, Peng Z, Wang X, Wang R, Huang YC, Meng F, Li M, Dong CL, Zhang Q, Gu L, Shen S. Anchoring single nickel atoms on carbon-vacant carbon nitride nanosheets for efficient photocatalytic hydrogen evolution. Chem Res Chin Univ. 2022;38(5):1243. https://doi.org/10.1007/s40242-022-2194-7.

[32] Wang W, Zhou H, Liu Y, Zhang S, Zhang Y, Wang G, Zhang H, Zhao H. Formation of B–N–C coordination to stabilize the exposed active nitrogen atoms in g-C_3N_4 for dramatically enhanced photocatalytic ammonia synthesis performance. Small. 2020;16(13):1906880. https://doi.org/10.1002/smll.201906880.

[33] Yang M, Lian R, Zhang X, Wang C, Cheng J, Wang X. Photocatalytic cyclization of nitrogen-centered radicals with carbon nitride through promoting substrate/catalyst interaction. Nat Commun. 2022;13(1):4900. https://doi.org/10.1038/s41467-022-32623-3.

[34] Ou H, Ning S, Zhu P, Chen S, Han A, Kang Q, Hu Z, Ye J, Wang D, Li Y. Carbon nitride photocatalysts with integrated oxidation and reduction atomic active centers for improved CO_2 conversion. Angew Chem Int Ed. 2022;61(34):e202206579. https://doi.org/10.1002/anie.202206579.

[35] Zhou AQ, Yang JM, Zhu XW, Zhu XL, Liu JY, Zhong K, Chen HX, Chu JY, Du YS, Song YH, Qian JC, Li HM, Xu H. Self-assembly construction of NiCo LDH/ultrathin g-C_3N_4 nanosheets photocatalyst for enhanced CO_2 reduction and charge separation mechanism study. Rare Met. 2022;41(6):2118. https://doi.org/10.1007/s12598-022-01960-z.

[36] Ou M, Wan S, Zhong Q, Zhang S, Song Y, Guo L, Cai W, Xu Y. Hierarchical Z-scheme photocatalyst of g-C_3N_4@Ag/$BiVO_4$(040) with enhanced visible-light-induced photocatalytic oxidation performance. Appl Catal B. 2018;221:97. https://doi.org/10.1016/j.apcatb.2017.09.005.

[37] Liu F, Shi R, Wang Z, Weng Y, Che CM, Chen Y. Direct Z-scheme hetero-phase junction of black/red phosphorus for photocatalytic water splitting. Angew Chem Int Ed. 2019;58(34):11791. https://doi.org/10.1002/anie.201906416.

[38] Zhou H, Wang M, Wang F. Oxygen-controlled photo-reforming of biopolyols to CO over Z-scheme CdS@g-C_3N_4. Chem. 2022;8(2):465. https://doi.org/10.1016/j.chempr.2021.10.021.

[39] Xu H, She X, Fei T, Song Y, Liu D, Li H, Yang X, Yang J, Li H, Song L, Ajayan PM, Wu J. Metal-oxide-mediated subtractive manufacturing of two-dimensional carbon nitride for high-efficiency and high-yield photocatalytic H_2 evolution. ACS Nano. 2019;13(10):11294. https://doi.org/10.1021/acsnano.9b04443.

[40] Meng J, Wang X, Liu Y, Ren M, Zhang X, Ding X, Guo Y, Yang Y. Acid-induced molecule self-assembly synthesis of Z--scheme WO_3/g-C_3N_4 heterojunctions for robust photocatalysis against phenolic pollutants. Chem Eng J. 2021;403:126354. https://doi.org/10.1016/j.cej.2020.126354.

[41] Liang ZY, Wei JX, Wang X, Yu Y, Xiao FX. Elegant Z--scheme-dictated g-C_3N_4 enwrapped WO_3 superstructures: a multifarious platform for versatile photoredox catalysis. J Mater Chem A. 2017;5(30):15601. https://doi.org/10.1039/C7TA04333C.

[42] Fu J, Xu Q, Low J, Jiang C, Yu J. Ultrathin 2D/2D WO_3/g-C_3N_4 step-scheme H_2-production photocatalyst. Appl Catal B. 2019;243:556. https://doi.org/10.1016/j.apcatb.2018.11.011.

[43] Chen X, Wang J, Chai Y, Zhang Z, Zhu Y. Efficient photocatalytic overall water splitting induced by the giant internal electric field of a g-C_3N_4/rGO/PDIP Z-scheme heterojunction. Adv Mater. 2021;33(7):2007479. https://doi.org/10.1002/adma.202007479.

[44] Zhao D, Dong CL, Wang B, Chen C, Huang YC, Diao Z, Li S, Guo L, Shen S. Synergy of dopants and defects in graphitic

18677185, 2024, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1007/s12598-024-02714-9 by Nanjing University Post And Telecommunications, Wiley Online Library on [13/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

carbon nitride with exceptionally modulated band structures for efficient photocatalytic oxygen evolution. Adv Mater. 2019; 31(43):1903545. https://doi.org/10.1002/adma.201903545.

[45] Zhao D, Wang Y, Dong CL, Huang YC, Chen J, Xue F, Shen S, Guo L. Boron-doped nitrogen-deficient carbon nitride-based Z-scheme heterostructures for photocatalytic overall water splitting. Nat Energy. 2021;6(4):388. https://doi.org/10.1038/s41560-021-00795-9.

[46] Zhao D, Wang Y, Dong CL, Meng F, Huang YC, Zhang Q, Gu L, Liu L, Shen S. Electron-deficient Zn-N_6 configuration enabling polymeric carbon nitride for visible-light photocatalytic overall water splitting. Nano-Micro Lett. 2022;14(1):223. https://doi.org/10.1007/s40820-022-00962-x.

[47] Wang Z, Li C, Domen K. Recent developments in heterogeneous photocatalysts for solar-driven overall water splitting. Chem Soc Rev. 2019;48(7):2109. https://doi.org/10.1039/C8CS00542G.

[48] Wang Q, Hisatomi T, Jia Q, Tokudome H, Zhong M, Wang C, Pan Z, Takata T, Nakabayashi M, Shibata N, Li Y, Sharp ID, Kudo A, Yamada T, Domen K. Scalable water splitting on particulate photocatalyst sheets with a solar-to-hydrogen energy conversion efficiency exceeding 1%. Nat Mater. 2016;15(6): 611. https://doi.org/10.1038/nmat4589.

[49] Song X, Li W, Liu X, Wu Y, He D, Ke Z, Cheng L, Jiang C, Wang G, Xiao X, Li Y. Oxygen vacancies enable the visible light photoactivity of chromium-implanted TiO_2 nanowires. J Energy Chem. 2021;55:154. https://doi.org/10.1016/j.jechem.2020.07.013.

[50] Kresse G, Furthmüller J. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. Comput Mater Sci. 1996;6(1):15. https://doi.org/10.1016/0927-0256(96)00008-0.

[51] Grimme S, Antony J, Ehrlich S, Krieg H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu. J Chem Phys. 2010;132(15):154104. https://doi.org/10.1063/1.3382344.

[52] Pawar RC, Kang S, Han H, Choi H, Lee CS. In situ reduction and exfoliation of g-C_3N_4 nanosheets with copious active sites via a thermal approach for effective water splitting. Catal Sci Technol. 2019;9(4):1004. https://doi.org/10.1039/C8CY02318B.

[53] Katsumata K, Motoyoshi R, Matsushita N, Okada K. Preparation of graphitic carbon nitride (g-C_3N_4)/WO_3 composites and enhanced visible-light-driven photodegradation of acetaldehyde gas. J Hazard Mater. 2013;260:475. https://doi.org/10.1016/j.jhazmat.2013.05.058.

[54] Zhang D, Guo Y, Zhao Z. Porous defect-modified graphitic carbon nitride via a facile one-step approach with significantly enhanced photocatalytic hydrogen evolution under visible light irradiation. Appl Catal B. 2018;226:1. https://doi.org/10.1016/j.apcatb.2017.12.044.

[55] Jin B, Yao G, Wang X, Ding K, Jin F. Photocatalytic oxidation of glucose into formate on nano TiO_2 catalyst. ACS Sustain Chem Eng. 2017;5(8):6377. https://doi.org/10.1021/acssuschemeng.7b00364.

[56] Yu S, Li J, Zhang Y, Li M, Dong F, Zhang T, Huang H. Local spatial charge separation and proton activation induced by surface hydroxylation promoting photocatalytic hydrogen evolution of polymeric carbon nitride. Nano Energy. 2018;50:383. https://doi.org/10.1016/j.nanoen.2018.05.053.

[57] Wang Y, Li X, Liu S, Liu Y, Kong T, Zhang H, Duan X, Chen C, Wang S. Roles of catalyst structure and gas surface reaction in the generation of hydroxyl radicals for photocatalytic oxidation. ACS Catal. 2022;12(5):2770. https://doi.org/10.1021/acscatal.1c05447.

[58] Li YY, Si Y, Zhou BX, Huang WQ, Hu W, Pan A, Fan X, Huang G-F. Strategy to boost catalytic activity of polymeric carbon nitride: synergistic effect of controllable in situ surface engineering and morphology. Nanoscale. 2019;11(35):16393. https://doi.org/10.1039/C9NR05413H.

[59] Coridan RH, Shaner M, Wiggenhorn C, Brunschwig BS, Lewis NS. Electrical and photoelectrochemical properties of WO_3/Si tandem photoelectrodes. J Phys Chem C. 2013;117(14):6949. https://doi.org/10.1021/jp311947x.

[60] Uekert T, Kasap H, Reisner E. Photoreforming of nonrecyclable plastic waste over a carbon nitride/nickel phosphide catalyst. J Am Chem Soc. 2019;141(38):15201. https://doi.org/10.1021/jacs.9b06872.

[61] Wang H, Yuan X, Wang H, Chen X, Wu Z, Jiang L, Xiong W, Zeng G. Facile synthesis of Sb_2S_3/ultrathin g-C_3N_4 sheets heterostructures embedded with g-C_3N_4 quantum dots with enhanced NIR-light photocatalytic performance. Appl Catal B. 2016;193:36. https://doi.org/10.1016/j.apcatb.2016.03.075.

[62] Lei B, Cui W, Chen P, Chen L, Li J, Dong F. C-doping induced oxygen-vacancy in WO_3 nanosheets for CO_2 activation and photoreduction. ACS Catal. 2022;12(15):9670. https://doi.org/10.1021/acscatal.2c02390.

[63] Lin Z, Wang Y, Peng Z, Huang YC, Meng F, Chen JL, Dong CL, Zhang Q, Wang R, Zhao D, Chen J, Gu L, Shen S. Single-metal atoms and ultra-small clusters manipulating charge carrier migration in polymeric perylene diimide for efficient photocatalytic oxygen production. Adv Energy Mater. 2022; 12(26):2200716. https://doi.org/10.1002/aenm.202200716.

[64] Li H, Yu H, Quan X, Chen S, Zhang Y. Uncovering the key role of the fermi level of the electron mediator in a Z-scheme photocatalyst by detecting the charge transfer process of WO_3-metal-gC_3N_4 (metal = Cu, Ag, Au). ACS Appl Mater Interfaces. 2016;8(3):2111. https://doi.org/10.1021/acsami.5b10613.

[65] Wang H, Sun X, Li D, Zhang X, Chen S, Shao W, Tian Y, Xie Y. Boosting hot-electron generation: exciton dissociation at the order-disorder interfaces in polymeric photocatalysts. J Am Chem Soc. 2017;139(6):2468. https://doi.org/10.1021/jacs.6b12878.

[66] Zhang Z, Yates JT Jr. Band bending in semiconductors: chemical and physical consequences at surfaces and interfaces. Chem Rev. 2012;112(10):5520. https://doi.org/10.1021/cr3000626.

[67] Bie C, Zhu B, Wang L, Yu H, Jiang C, Chen T, Yu J. A bifunctional CdS/MoO_2/MoS_2 catalyst enhances photocatalytic H_2 evolution and pyruvic acid synthesis. Angew Chem Int Ed. 2022;61(44):e202212045. https://doi.org/10.1002/anie.202212045.

[68] Liu K, Ma J, Yang X, Liu Z, Li X, Zhang J, Cui R, Sun R. Phosphorus/oxygen co-doping in hollow-tube-shaped carbon nitride for efficient simultaneous visible-light-driven water splitting and biorefinery. Chem Eng J. 2022;437:135232. https://doi.org/10.1016/j.cej.2022.135232.

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